

2.1.1 Roots of Quadratic Equation

ALGORITHM

1. Start
2. Import the math library.
3. Read three integers a, b, and c (coefficients of the quadratic equation).
4. Calculate the discriminant

$$D = b^2 - 4ac$$

5. If $D > 0$:

Calculate two real and different roots using:

$$\frac{-b+\sqrt{D}}{2a}, \frac{-b-\sqrt{D}}{2a}$$

Print both roots up to 2 decimal places.

- 6 Else if $D == 0$:

Calculate the single repeated root:

$$\frac{-b}{2a}$$

Print the root twice up to 2 decimal places.

- 7 Else ($D < 0$):

Calculate real part:

$$\frac{-b}{2a}$$

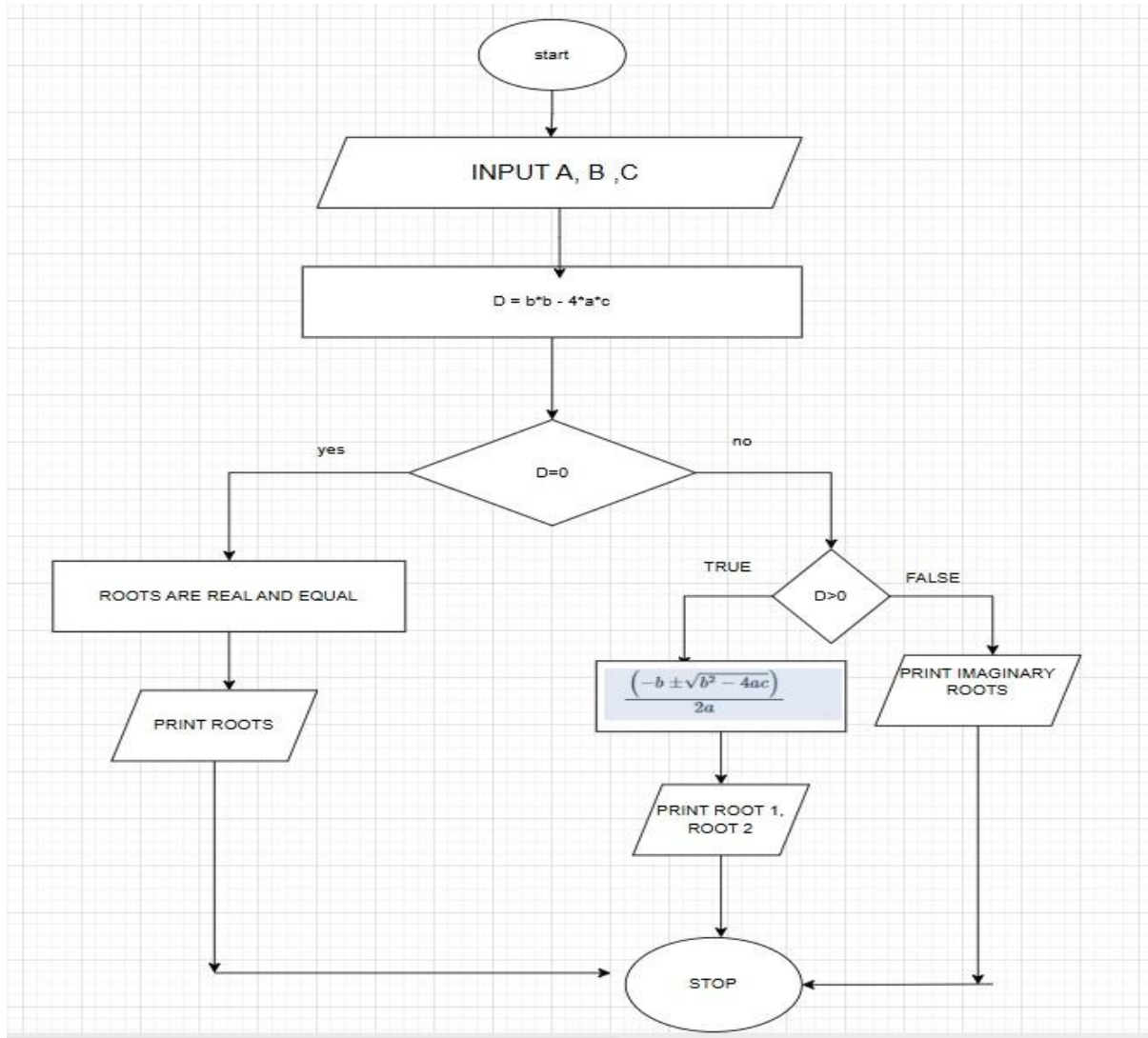
Calculate imaginary part:

$$\frac{\sqrt{-D}}{2a}$$

Print both complex roots up to 2 decimal places.

- 8 Stop

FLOWCHART



PROGRAM

2.1.1. Roots of a Quadratic Equation

Write a program to find the roots of a quadratic equation, given its coefficients a , b , and c . Use the quadratic formula:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a , b , and c , respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <root1>
root2 = <root2>
```

- If roots are the same, print:

```
root1 = root2 = <root1>
```

- If roots are imaginary, print:

```
root1 = <realPart>+<imaginaryPart>i
root2 = <realPart>-<imaginaryPart>i
```

All values should be formatted to two decimal places.

Sample Test Cases

quadratic...

```

1 # write your code here
2 a, b, c = map(float, input().split())
3 D = (b*b) - (4*a*c)
4 sqrtD = D ** 0.5
5 root1 = (-b+sqrtD)/(2*a)
6 root2 = (-b-sqrtD)/(2*a)
7 if D > 0:
8     print(f"root1 = {root1:.2f}")
9     print(f"root2 = {root2:.2f}")
10 elif D == 0:
11     print(f"root1 = root2 = {root1:.2f}")
12 else:
13     print(f"root1 = {root1.real:.2f}{root1.imag:+.2f}i")
14     print(f"root2 = {root2.real:.2f}{root2.imag:+.2f}i")
  
```

Average time: 0.003 s, Maximum time: 0.005 s

3 out of 3 shown test case(s) passed, 3 out of 3 hidden test case(s) passed

Test case 1: Expected output: 1 -5.6, Actual output: 1 -5.6

Test case 2: Expected output: root1 = 3.00, root2 = 2.00, Actual output: root1 = 3.00, root2 = 2.00